Field Notes on Coastal Erosion Monitoring

Coastal erosion is measured by comparing shoreline positions over time. Our team surveyed four beaches between March and October, using fixed markers and satellite positioning.

The northern beach lost an average of 1.8 metres of width, mostly after two storms in late September. The southern beaches were stable, protected by a natural reef that dissipates wave energy.

Sediment samples were collected at low tide along transects perpendicular to the shore. Grain size analysis showed coarser sand where wave action was strongest.

Volunteers from local schools helped record observations, which improved coverage on weekends and holidays when professional staff were not available.

We recommend repeating the survey every six months and adding drone photogrammetry to capture changes in dune height that ground measurements may miss.

All raw measurements are archived with their timestamps so that future studies can reproduce the analysis and extend the time series.

Beach	Width change (m)	Storm events
North	-1.8	2
Central	-0.6	1
South A	0.1	0
South B	0.0	0